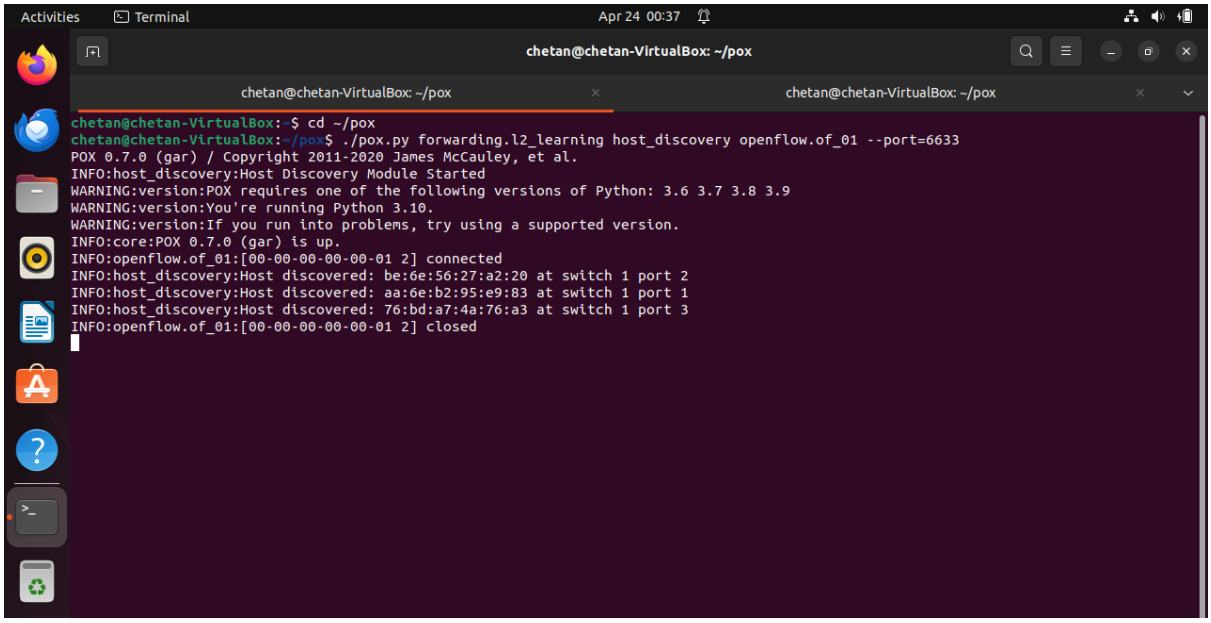
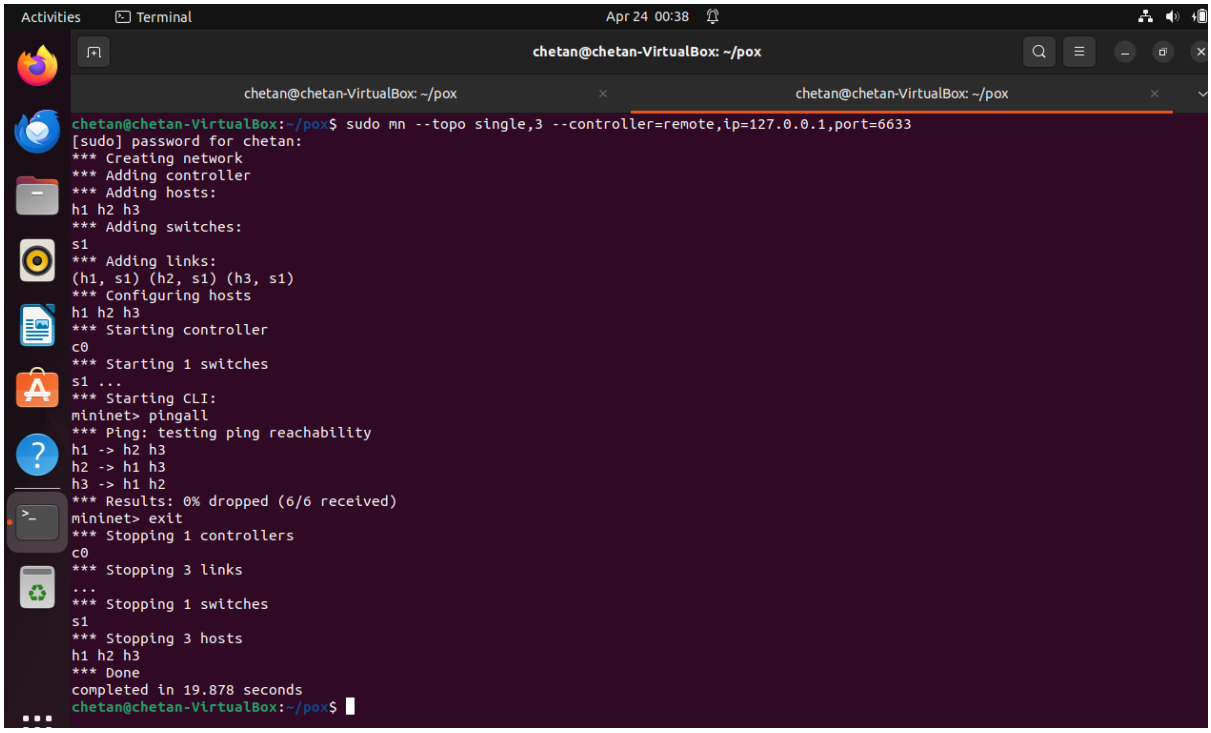
CHETAN , SECTION C, SRN: PES2UG24CS131 , SDN REPORT

TOPIC: HOST DISCOVERY SERVICE

AUTOMATICALLY DETECT AND MAINTAIN A LIST OF HOSTS IN THE SDN NETWORK

DETECT HOST JOIN EVENTS, MAINTAIN HOST DETAILS, UPDATE DYNAMICALLY





This project demonstrates Software Defined Networking using Mininet and POX controller.
The control plane is separated from the data plane, and the controller manages packet forwarding.

./pox.py forwarding.l2_learning host_discovery openflow.of_01 --port=6633

The controller is now running and detecting hosts dynamically.

We can see that hosts are discovered on different ports of the switch.

sudo mn --topo single,3 --controller=remote,ip=127.0.0.1,port=6633

Now I create a topology with one switch and three hosts connected to a remote controller.

This is a single switch topology where h1, h2, and h3 are connected to switch s1, which is controlled by POX.

```
sudo mn --topo single,3 --controller=remote,ip=127.0.0.1,port=6633
```

Now I create a topology with one switch and three hosts connected to a remote controller.

Results: 0% dropped (6/6 received)

All hosts are successfully communicating, which shows that the controller is correctly managing the network.

Thus, this project demonstrates centralized control and dynamic host discovery using SDN.